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RESEARCH ARTICLE

Enhanced Weighted Target Approximation Using an Improved Particle Swarm Optimization Approach

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ABSTRACT This article proposes a two-layer tactical decision architecture for the autonomous allocation of Autonomous Aerial Vehicle (AAV) swarms in Search-and-Rescue (SAR) missions under severe resource constraints. The approach addresses the triage and distributed delivery challenge by integrating a *Weighted Target Approximation* protocol enhanced by Particle Swarm Optimization (WTA–PSO) with a Spatial Tactical Evaluation heuristic (*Spatial Clustering*). In the first layer, WTA–PSO replaces the conventional proximity-based target choice with a low-cost one-dimensional optimization that jointly accounts for flight distance, the victim's medical urgency, and local drone-to-drone conflict, while preserving WTA's intention exchange and distributed conflict resolution. In the second layer, spatial clustering enables AAVs, each constrained to a single rescue payload (*single-payload drop*) to evaluate the disaster topology and focus on victim clusters that lie within the payload coverage radius. Experiments were conducted in a dynamic two-dimensional simulation environment with mobile targets categorized into three urgency levels (*green*, *orange*, *gray*). The results reveal complementary gain profiles: under strict one-to-one engagement, WTA–PSO significantly increases the global Humanitarian Impact (*Total_Value*) exclusively through intelligent prioritization (emphasizing the critical *gray* level), while keeping the service volume near the fleet's physical limit. However, when spatial clustering is enabled (Layer 2), swarm intelligence enables multiple simultaneous rescues with a single payload, breaking the logistical ceiling imposed by the fleet size (N_{uav}) and dramatically maximizing mission efficiency. The architecture's robustness is validated over 50 independent runs (*seeds*) with high statistical significance (Kruskal–Wallis and paired Wilcoxon tests, $p < 0.01$), along with scalability analyses for reduced ($N_{\mathrm{uav}} = 15$) and supersaturated ($N_{\mathrm{uav}} = 60$) fleets. The findings show that coupling WTA–PSO's predictive conflict-mitigation capability with the geometric exploitation of victim distributions consistently maximizes the operational effectiveness of robotic swarms across disasters of different scales.

INDEX TERMS Drone swarm, search and rescue (SAR), WTA–PSO, particle swarm optimization, task allocation, CACOC.

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I. INTRODUCTION

In disaster scenarios, the narrow survival window of victims demands rapid response and stringent mitigation of operational risks. To address these critical demands in complex environments, the adoption of autonomous robotic systems has become established as a fundamental solution, enabling more efficient and real-time coordinated Search and Rescue (SAR) operations [1]. In this context, given the frequent collapse of terrestrial communication infrastructure, the rapid deployment of swarms of Autonomous Aerial Vehicles (AAVs) serves as an immediate and decentralized support network [2]. By operating cooperatively, these fleets restore tactical coverage and overcome the limitations of isolated vehicles, rapidly adapting to the unpredictable dynamics of disaster zones [2]. This decentralized architecture proves crucial in remote and topographically inaccessible regions, where swarm intelligence ensures mission continuity and rapid victim detection even in the absence of central control or GNSS signals [3], [28]. Through adaptive mesh networks, these fleets maintain operational resilience and drastically reduce search time compared with conventional methods and vehicles [3]. However, the full effectiveness of these operations is constrained by severe real-world logistical and operational bottlenecks [4]. Recent literature indicates that the lack of efficient methods for task scheduling and resource optimization under strict communication and payload constraints remains a central challenge in coordinating these swarms [4]. To overcome this obstacle, the capability for dynamic target prioritization becomes imperative. Recent studies show that tactical decision-making guided by different priority levels is essential to optimize swarm positioning when simultaneously servicing multiple targets [5]. Thus, integrating triage mechanisms into task planning emerges as a crucial step toward managing the trade-off between search speed and the efficient allocation of scarce resources [5].

Despite the urgency inherent to disaster scenarios, autonomous fleets often operate under a form of "spatial myopia," making allocation decisions guided strictly by geometric proximity. This exclusive reliance on distance introduces a critical operational bias, in which severely injured victims may be neglected, highlighting that task planning must explicitly incorporate priority-awareness mechanisms [6]. Beyond this prioritization failure, the logistical impact of the swarm is further constrained by dispatch inefficiencies. Recent literature shows that strict payload limitations directly affect the time and cost of operations, requiring allocation strategies that minimize excessive vehicle usage in order to prevent premature fleet exhaustion in complex scenarios [7]. To enable the scaling of operations under such constraints, the transition from strictly individual allocations to clustering- and load-aware methods has proven essential. By organizing the swarm into cooperative groups, it becomes possible to distribute demands more efficiently and avoid the critical overloading of isolated vehicles [8].

To overcome these operational limitations, this article proposes a two-layer tactical architecture for AAV swarms. Extending the concept of value-driven decentralized coordination [9], the first layer introduces the WTA-PSO (Weighted Target Approximation) protocol. By incorporating medical triage directly into intention generation, this approach eliminates "spatial myopia" and ensures the real-time prioritization of critical victims at low computational cost. Complementarily, the second layer of the architecture introduces a clustering-based Spatial Tactical Assessment heuristic. This approach overcomes the logistical limitation of individual engagement by consolidating geographically proximate demands. By allowing a single AAV to service a cluster of victims through area-effect deliveries, the method prevents premature vehicle exhaustion, thereby ensuring operational scalability and preserving the fleet's response capability in high-density zones.

The remainder of this article is organized as follows. Section II describes the Materials and Methods, detailing the proposed two-layer decision architecture, including the mathematical formulation of the WTA-PSO protocol and the spatial clustering heuristic, as well as the design of the simulation environment. Section III presents the Results obtained, highlighting the tactical performance of the fleet and the corresponding statistical validation under different density scales. Section IV provides the Discussion, analyzing the logistical impact of the architecture in light of real-world SAR mission bottlenecks. Finally, Section V presents the Conclusion of the study, synthesizing the contributions and outlining directions for future work.

A. LITERATURE REVIEW

The integration of Autonomous Aerial Vehicles (AAVs) with Artificial Intelligence algorithms has shown strong potential for improving the efficiency of disaster response operations. Recent reviews highlight the use of AAVs in Search and Rescue (*Search and Rescue* – SAR) missions, emphasizing their ability to rapidly cover large areas, locate victims, and deliver first-aid supplies, especially in hard-to-reach regions and obstacle avoidance [10], [26], [27], [30]. Despite this progress, the transition toward fully autonomous swarms still faces logistical and coordination challenges. In this context, Multi-Robot Task Allocation (MRTA) constitutes a complex problem, since the environment evolves throughout the mission and new targets may emerge during execution. Recent studies show that centralized approaches, although effective in static and pre-planned scenarios, become increasingly limited as fleet size and the number of demands grow, due to increased latency and communication overload at the central node [11]. For this reason, in dynamic SAR scenarios, distributed architectures have been regarded as more suitable, as they enable faster local responses and greater robustness against single-point failures [11]. Within this paradigm, *market-based* solutions, rooted in foundational utility-driven negotiation models [12], and optimization methods have been widely investigated because of their scalability and ability to adapt to unpredictable environmental changes [13].

Even so, recent reviews indicate that the literature still lacks sufficiently robust multi-objective strategies to handle rapid task reallocation under severe communication constraints and physical fleet limitations [13]. In the absence of more agile decision-making mechanisms, allocation often relies on simple heuristics based solely on geometric distance, which may lead to the improper prioritization of less critical targets. Thus, the question of which algorithmic approach can combine low computational cost with real-time tactical decision-making capability for onboard execution on AAVs remains open.

To address the limitations of these heuristics, the literature has extensively explored bio-inspired algorithms for the dynamic coordination of multiple AAVs [14]. Although Genetic Algorithms (GAs) are effective for *offline* global optimization, their *on-the-fly* tactical application is limited by the computational cost associated with encoding and mutation operations [15]. This cost may compromise decisions that must be made within fractions of a second on board the drones. In contrast, swarm intelligence methods, particularly Particle Swarm Optimization (PSO), have proven more suitable for resource-constrained *hardware*, offering greater agility than classical evolutionary methods [14], [25]. Originally proposed by Kennedy and Eberhart, PSO achieves this lightweight behavior because it does not depend on the iterative regeneration of entire populations [16]. Instead, the algorithm updates particle velocity and position through simple mathematical and stochastic operations, which favors rapid convergence at low computational cost [16]. Recent comparative studies in SAR missions reinforce this efficiency, indicating that PSO offers a favorable balance between tactical performance and spatial exploration of the environment [17]. However, the state of the art also points to an important limitation: the standard formulation of the algorithm was designed for static targets and presents convergence and coordination difficulties in probabilistic scenarios, in which victims move or the environment changes rapidly [17]. In response, more recent studies have proposed enhanced PSO variants, adjusting its formulation to improve convergence in complex AAV routing scenarios [18]. Based on this gap, this study introduces the WTA-PSO architecture, combining multi-objective triage assessment with a high-speed iterative micro-swarm to prioritize critical victims in real time.

Beyond the convergence speed of each algorithm, the tactical effectiveness of an autonomous swarm also depends on its ability to resolve conflicts and avoid redundant allocations in a decentralized manner. In this setting, distributed negotiation mechanisms and utility-based protocols have been identified as effective alternatives. Recent studies show that allocation methods driven by the swarm's dynamic utility, simultaneously considering variables such as distance and task criticality, provide coordination that is more adaptable to environmental changes [19]. Building on this value-driven coordination perspective, previous work demonstrated the effectiveness of distributed communication in dynamic environments [9]. By modeling the swarm with coevolutionary genetic algorithms (CompCGA), it was observed that interaction among agents is an important factor in improving the quality of tactical allocation: although it does not significantly increase the total number of interceptions, decentralized negotiation promotes an emergent spatial adaptation that prioritizes targets of higher strategic value [9].

However, the generation of optimized intentions and target prioritization addresses only part of the problem, since logistical limitations of the classical rescue model still remain. Recent assessments show that a large portion of MRTA architectures for disaster scenarios still adopts the *Single-Task* (ST) paradigm, formally classified in classical coordination taxonomies as ST-SR (*Single-Task robots*, *Single-Robot tasks*) [20], in which each vehicle is assigned to a single task or victim [21]. Under conditions of severe resource scarcity, this one-to-one relationship may lead to rapid fleet exhaustion and reduced efficiency in the presence of dense clusters of simultaneous emergencies. To address this limitation, recent studies have proposed *clustering-based task allocation* methods. By segmenting the environment and grouping geographically proximate tasks, these approaches allow a single vehicle to serve multiple targets within the same cluster, overcoming the one-to-one restriction and expanding the fleet's logistical impact [22]. It is in this context that this work proposes a two-layer architecture. The approach integrates the stochastic intention generation of the WTA-PSO protocol (Layer 1) with an elastic-constraint Spatial Clustering heuristic (Layer 2), seeking to move from isolated point allocation to area-effect tactical engagement, while supporting critical victims even under severe fleet underrepresentation conditions.

II. METHODOLOGY

This study extends the coevolutionary simulation framework from a previous work [9], adapting it to the Search and Rescue (SAR) context by introducing a two-layer decision architecture within the communication protocol. The foundational components employ literature-inspired models, such as the predator-prey dynamics by Stolfi et al. [23]. The specific formulations and parameterizations adopted in this framework (AAV mobility via CACOC, victim drift behavior via P-MoM, and competitive coevolution via CompCGA) follow those established in our previous work [9], and are described herein to the extent necessary for reproducibility. The methodological contribution of this paper encompasses: (i) the WTA-PSO mechanism, which solely alters the local candidate selection rule at the moment of generating the intention, and (ii) the Spatial Tactical Evaluation (*Spatial Clustering*), which enables area engagements (*Area-of-Effect*). The WTA stages related to intention exchange, distributed conflict resolution, priority tie-breaking, and commitment remain unchanged.

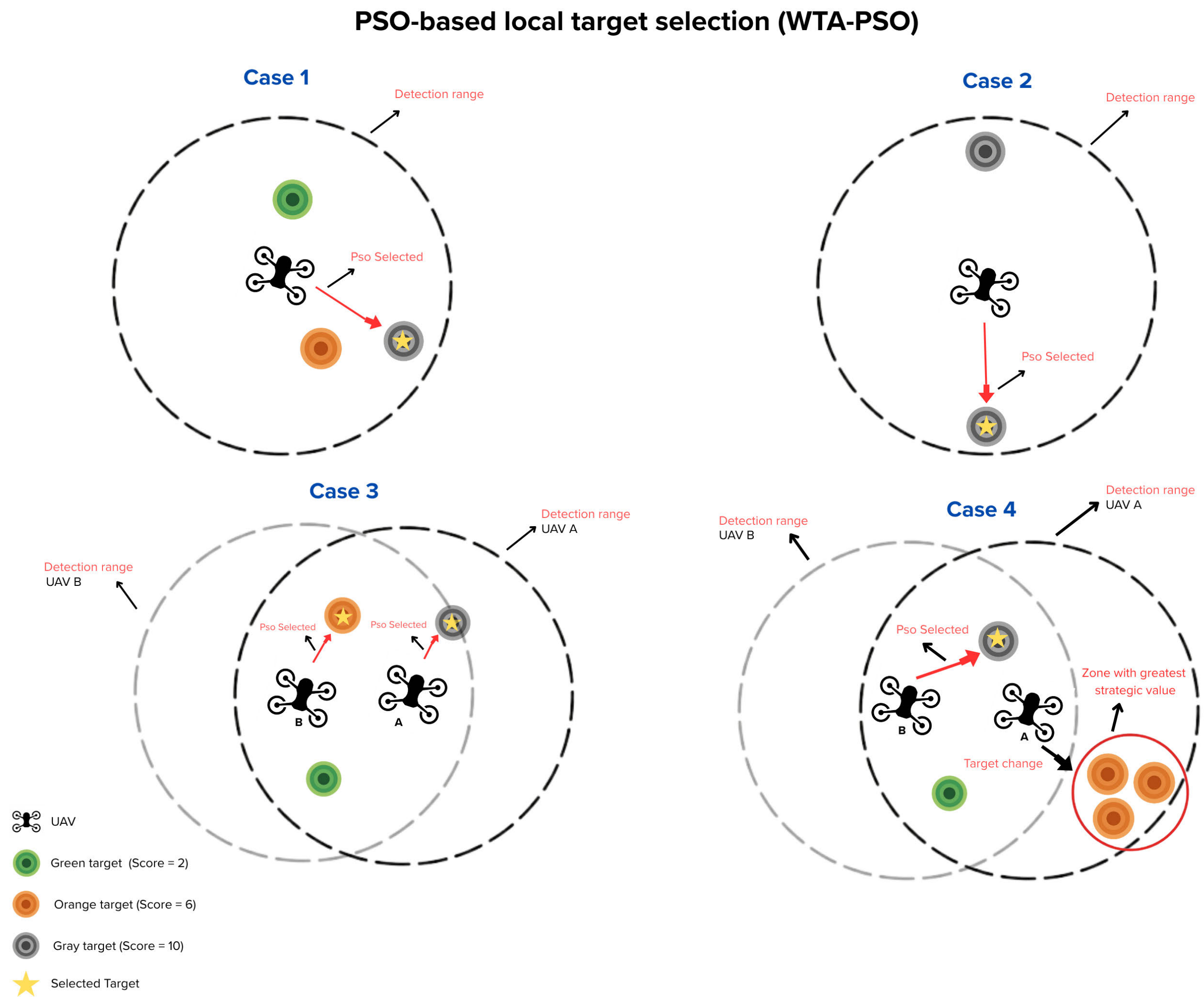

FIGURE 1. PSO-based local target selection (WTA-PSO). Cases 1 to 3 illustrate Layer 1 mechanics, including urgency-based selection, stochastic tie-breaking, and predictive conflict mitigation. Case 4 demonstrates the Layer 2 Spatial Tactical Evaluation, where AAVs switch trajectories to maximize the rescue impact within a drop zone.

A. SIMULATION ENVIRONMENT

A two-dimensional disaster environment is simulated consisting of 100 × 100 cells, where 1 cell corresponds to 100 m, resulting in an approximate area of 10 × 10 km. Agent positions are represented by continuous 2D coordinates and updated using periodic boundary conditions (*wrap-around*) within the simulation space. In each run, the initial positions are randomly sampled, and the evolutionary process is executed for 50 generations. In every generation, fitness is estimated from a simulation episode with a fixed number of internal iterations (Section II-D).

The complete execution loop of these internal iterations is illustrated in Figure 2. The sequence begins with the swarm deployment and parameter initialization, followed by a continuous monitoring loop. In each step, the network state is synchronized, and the AAV mobility models are applied. Upon detecting a potential target, the system triggers the local target selection phase, using either a proximity baseline or the PSO approach, followed by the WTA protocol to resolve conflicts and commit to the rescue. This pursuit cycle repeats until the fleet is exhausted or the maximum number of iterations is reached.

B. RESCUE FLEET AND VICTIM MODEL (AAV-VICTIM MODEL)

The environment comprises two classes of agents: (i) rescue AAVs and (ii) victims (mobile agents undergoing displacement/drift). The AAVs aim to detect and rescue victims, whereas the victims' behavior simulates the increasing difficulty of localization. Both populations are optimized by a competitive coevolutionary genetic algorithm (CompCGA), consistent with [9].

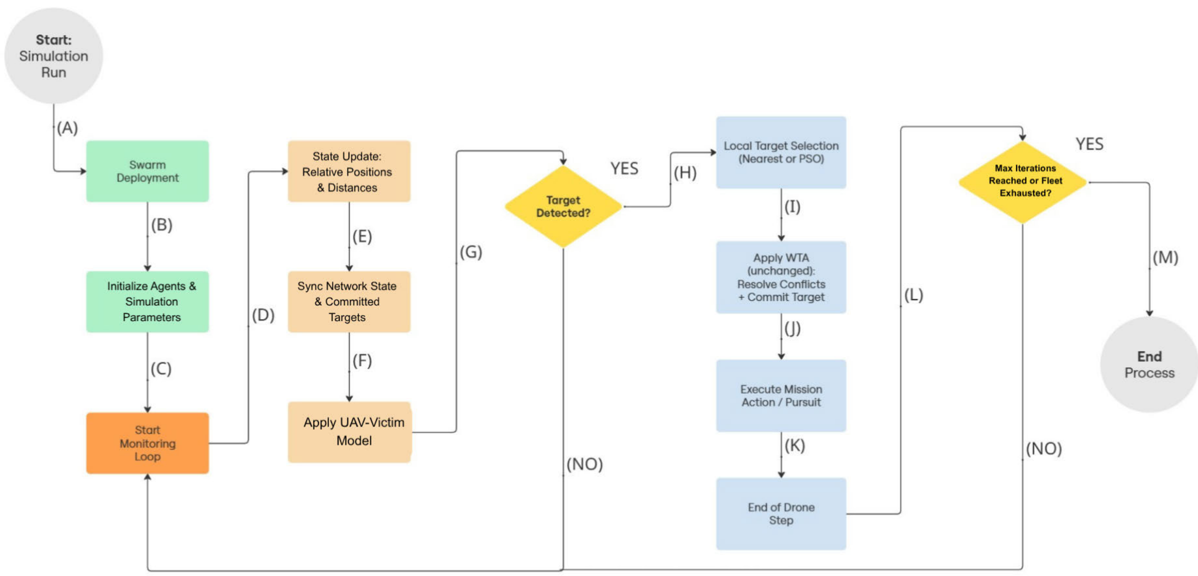

FIGURE 2. Flowchart of the WTA–PSO decision loop.

1) AAV MOBILITY MODEL (CACOC)

The AAVs follow a mobility model based on CACOC (*Chaotic Ant Colony Optimization for Coverage*), an adaptation of chaotic coverage strategies inspired by [23]. The model is driven by a chaotic return map and pheromone trail interactions, with implementation and parameterization consistent with a previous work [9]. Each AAV is parameterized by: trail adhesion amount (τ_a), repulsion radius (τ_r), and detection radius (τ_d), as summarized in Table 1. When no victim is visible, the AAVs execute chaotic exploration influenced by the local trail density; when a target is selected, the AAV tends to direct its movement toward the corresponding victim, applying short-range repulsion to avoid collisions with other vehicles.

TABLE 1. Parameters of the rescue AAVs in the CACOC variant (*1 cell = 100 m*).

Parameter	Symbol	Unit	Initial range/value	Mutation (GA)
Coordinated trail adhesion	τ_a	–	(0.5–1.5)	± 0.05
Sweep repulsion	τ_r	cells	(3.0–8.0)	± 0.5
Victim detection radius	τ_d	cells	25	± 1.0

A rescue (supply delivery) is registered when the AAV reaches a distance less than or equal to the effective radius $r_c = 2.0$ (in cell units) from the victim. To model severe logistical constraints, following the delivery, both the AAV and the assisted victims are removed from the episode, consistent with the resource constraint mechanics of [9]. Consequently, each AAV can perform at most one delivery per episode, imposing an upper limit of N_{uav} deliveries per generation. If the rescue is not accomplished within a fixed horizon (15 internal iterations), the AAV discards the target and restarts the selection process.

2) DYNAMIC VICTIM BEHAVIOR MODEL (P-MoM)

The victims follow the P-MoM model [9], adapted to simulate erratic behavior and drift in disaster areas by combining relative dispersion, obstacle (trail) evasion, and inertial exploration. Each victim is parameterized by: dispersion weight (t_b), displacement persistence (t_c), and perception radius (t_d). The movement direction results from a weighted sum:

$$\vec{F}_{\text{total}} = t_b \cdot \vec{F}_{\text{dispersion}} + 0.3 \cdot \vec{F}_{\text{environment}} + t_c \cdot \vec{F}_{\text{exploration}}, \tag{1}$$

followed by normalization and the application of a fixed step size, as in [9].

TABLE 2. Victim displacement parameters in the coevolutionary model.

Parameter	Symbol	Unit	Initial range/value	Mutation (GA)
Initial position	pos_ini	cells	(10,10)–(90,90)	–
Relative dispersion weight	t_b	–	(0.5–1.0)	± 0.05
Displacement weight	t_c	–	class-dependent	–
Perception radius	t_d	cells	(5.0–10.0)	± 0.5

C. MEDICAL URGENCY LEVELS, QUANTITIES, AND SCORES

To model triage, the victims are grouped into three abstract classes identified by color and priority: *green* (low risk), *orange* (medium risk), and *gray* (critical condition). Scores of (2, 6, 10) are assigned to (*green*, *orange*, *gray*), respectively. The displacement weights (t_c) are class-dependent, following the same pattern as the base framework [9].

Three different scenario scales are adapted, keeping the map (100×100), the class scores, and the coordination rules fixed: (i) reduced scenario: $N_{\text{uav}} = 15$ and $N_{\text{victim}} = 85$

(50 *green*, 20 *orange*, 15 *gray*); (ii) baseline scenario: $N_{\text{uav}} = 30$ and $N_{\text{victim}} = 170$ (100 *green*, 40 *orange*, 30 *gray*); (iii) expanded scenario: $N_{\text{uav}} = 60$ and $N_{\text{victim}} = 340$ (200 *green*, 80 *orange*, 60 *gray*).

TABLE 3. Evaluated scenarios (scale), with quantities of AAVs and victims per class.

Scenario	N_{uav}	N_{victim}	Green	Orange	Gray
Reduced (0.5×)	15	85	50	20	15
Baseline (1×)	30	170	100	40	30
Expanded (2×)	60	340	200	80	60

D. COMPETITIVE COEVOLUTIONARY GENETIC ALGORITHM (COMPCGA)

A competitive coevolution architecture similar to that proposed in [9] is adopted. Two genetic algorithms run in parallel: GA_{uav} maximizes the fleet's performance (priority-weighted rescue impact), and GA_{victim} maximizes the difficulty of locating the victims. The parameterization and rules follow the previous work [9], employing elitist selection (AAVs: top 40%; victims: top 20% per class). Crossover utilizes real-coded interpolation, and mutation applies perturbations according to Tables 1 and 2. Each evolutionary generation is evaluated through a simulation episode of 30 internal iterations. Every 10 generations, the best individuals from each population are incorporated into the opposing population's evaluation set to maintain coevolutionary pressure, exactly as validated in [9].

E. COMMUNICATION AND TARGET ALLOCATION STRATEGY

Communication is utilized to reduce redundancies and improve allocation to higher-priority victims. Every three internal iterations, each AAV transmits an intention message containing (*victim_id*, distance, victim urgency). Target conflicts are resolved in a distributed manner: the AAV with the shortest distance wins; ties are broken by priority (*gray* > *orange* > *green*). Once confirmed, the victim's identifier is added to a shared list of committed targets.

1) BASELINE WTA (UNCHANGED CONFLICT RESOLUTION)

The WTA protocol from the base framework [9] is utilized without modifications in the conflict resolution, tie-breaking, and commitment stages. The only stage where the WTA–PSO intervenes is in the local target candidate selection, immediately prior to sending the intention. The re-evaluation logic during the approach is also preserved, following the criteria of the base framework [9].

2) LAYER 1: WTA–PSO (LOCAL SELECTION VIA PSO)

a: CORE IDEA

In the baseline WTA, the target candidate is selected in a strictly greedy and deterministic manner based on the shortest distance metric. In the WTA-PSO, this choice is replaced with a lightweight one-dimensional PSO Kennedy and Eberhart [16] that locally optimizes the candidate selection at the moment of intention generation. The adoption of a metaheuristic rather than a simple deterministic exhaustive search (arg min) does not aim to reduce the computational search cost, but rather to inject controlled stochasticity into the tactical decision. The particle dynamics allow one to break perfect spatial symmetries, avoiding impasses (*deadlocks*), drastically reducing the probability of multiple AAVs in the same region simultaneously converging on the exact same target even before the conflict resolution phase.

b: REPRESENTATION AND DISCRETE MAPPING

Let $\mathcal{V}_{vis}$ be the set of visible victims and $n = |\mathcal{V}_{vis}|$. Each particle has a continuous position $x \in [0, n-1]$, which is mapped to a discrete index $j \in \{0, 1, \ldots, n-1\}$ by rounding to the nearest integer: $j = \lfloor x \rceil$. The resulting index j directly identifies a candidate victim in $\mathcal{V}_{vis}$.

c: OBJECTIVE FUNCTION

To evaluate each candidate victim, a specific multi-objective cost function J is formulated, which is minimized by the standard PSO algorithm [16]:

$$J = \alpha \cdot dist_{norm} - \beta \cdot score_{norm} + \gamma \cdot conflict, \tag{2}$$

where α, β, and γ are the weighting coefficients for distance, urgency score, and conflict penalty, respectively. The normalized variables are defined as $dist_{norm} = \dfrac{d(uav, victim)}{\tau_d}$ and $score_{norm} = \left(\dfrac{score(victim)}{10}\right)^2$. The quadratic scaling of the urgency score ensures that critical victims (*gray*) are heavily prioritized even when moderately more distant than low-priority targets (*green*), reflecting the medical triage principle that life-threatening conditions demand immediate attention regardless of marginal accessibility differences. The *conflict* penalty is formally defined to mitigate local redundancy and avoid already engaged targets:

$$conflict(victim) = \frac{\sum_{u \in U_{other}} \mathbb{I}(int(u) = victim)}{\max(1, |U_{other}|)} + \rho \cdot \mathbb{I}(victim \in \mathcal{V}_{committed}) \tag{3}$$

where U_{other} represents the set of other active AAVs, $int(u)$ is the target of the current intention transmitted by AAV u, and $\mathbb{I}(\cdot)$ is the indicator function. Furthermore, $\mathcal{V}_{committed}$ is the shared set of targets already confirmed via WTA, and ρ is a safeguard penalty ($\rho = 1.0$). The use of the $\max(1, |U_{other}|)$ function in the denominator prevents division by zero in edge cases where a AAV operates entirely alone or is the last active agent in the network.

The practical effects of this objective function in Layer 1 are illustrated in Cases 1 to 3 of Figure 1. As shown in Case 1, medical urgency drives the selection: when multiple victims are within the detection range, the local WTA-PSO optimization directs the rescue AAV to the critical target

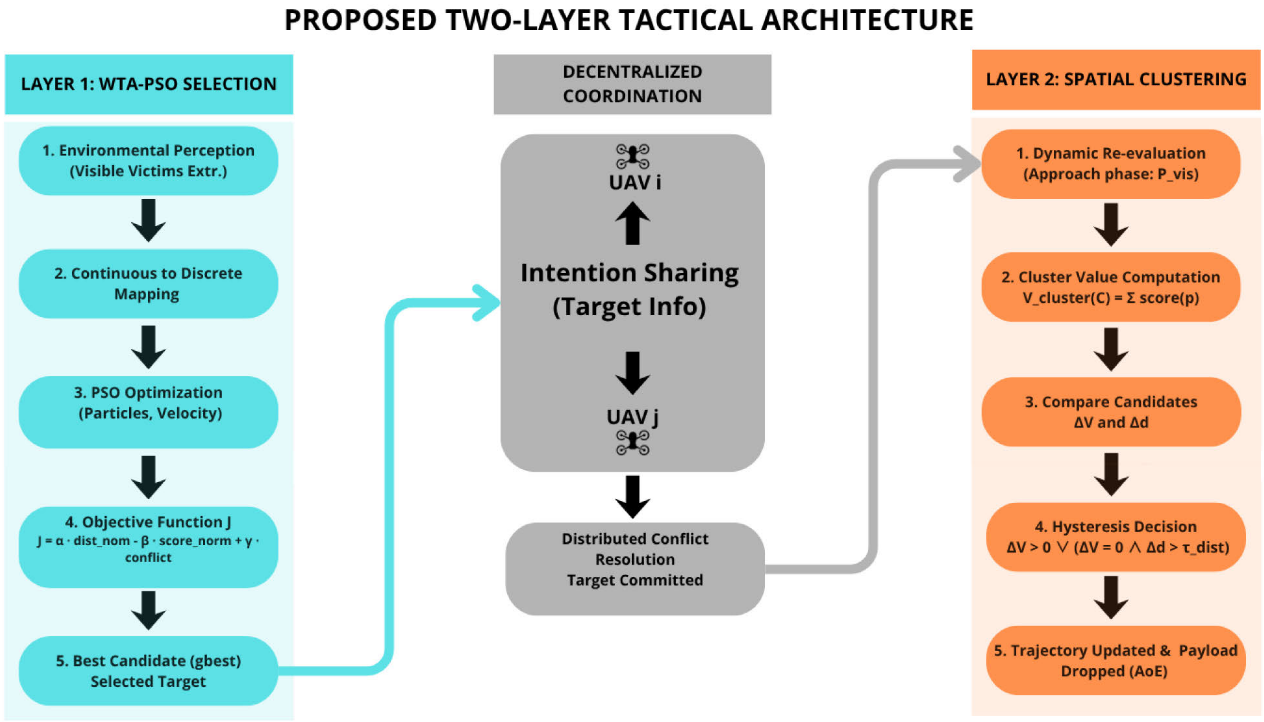

FIGURE 3. **Proposed tactical decision architecture for AAV swarms. The diagram illustrates the integration between intention optimization via PSO (Layer 1), distributed conflict resolution by distance and urgency, and the spatial clustering mechanism (Layer 2), which enables the system to overcome logistical constraints through simultaneous area-of-effect rescues.**

(*gray*), bypassing lower-risk victims (*green* and *orange*) even if they are spatially closer. Case 2 demonstrates the stochastic resolution of strict ties. If multiple visible victims present the exact same urgency level and distance, the cost function yields identical values. In these scenarios of perfect symmetry, the stochastic nature of the particle initialization guarantees a non-deterministic tie-breaking, preventing navigation deadlocks. Finally, Case 3 illustrates predictive conflict mitigation. In areas with sensory overlap, the tactical cost function penalizes competition ($\gamma \cdot conflict$). Consequently, the AAVs efficiently distribute themselves among the most valuable available targets (*gray* and *orange*), avoiding approach redundancy.

d: PARAMETERS AND COMPUTATIONAL COST

A dynamic budget is utilized to adapt the swarm size to the number of visible victims (n) is used to maximize real-time efficiency, capped at maximum limits for particles and iterations. An inertia weight with linear decay is employed [17], alongside velocity clamping to mitigate instabilities [17], [24]. Furthermore, an early stopping criterion is adopted if the global best value (*gbest*) stagnates for a predefined fraction of the total iterations. The specific numerical values for the PSO hyperparameters and the cost function weights (α, β, γ) are detailed in the experimental results (Section III-A). The complete intention generation logic is outlined in Algorithm 1.

e: INTEGRATION WITH THE WTA

After selecting *best* $\leftarrow$ PSO_Select_Victim(·), the AAV transmits its intention and applies the same WTA from the baseline framework [9]. Thus, the WTA–PSO alters only the local selection stage at the exact moment of intention generation, thereby preserving the comparability between methods.

3) LAYER 2: SPATIAL TACTICAL EVALUATION (SPATIAL TARGET CLUSTERING)

The effective operation of AAVs in disaster scenarios, where multiple victims may be spatially clustered, is considered by introducing the second layer to the decision architecture: the Spatial Tactical Evaluation (*Spatial Clustering*). Unlike strict individual engagement, in this stage, the AAV evaluates the local topology to perform a single supply delivery (*single-payload drop*) that maximizes the humanitarian rescue impact within a predefined area of effect.

During the approach flight, the AAV dynamically re-evaluates the subset of visible victims within its detection radius, denoted by P_{vis}. The tactical value of a cluster, anchored on a primary target C, is defined by the sum of the medical urgency scores (*triage score*) of all valid victims contained within the supply coverage zone. This formulation is given by Equation 4:

$$V_{cluster}(C) = \sum_{p \in P_{vis}} \text{score}(p) \cdot \mathbb{I}(d(C, p) \leq r_{drop}) \tag{4}$$

Algorithm 1 PSO Select Victim: Local Target Selection in WTA-PSO

Require: uav, $visible\_victims$, $other\_uavs$, $committed\_targets$

$n \leftarrow |visible\_victims|$
$P \leftarrow \min(20, \max(2, n))$ ▷ Dynamic swarm sizing
$T \leftarrow \min(15, \max(3, 2n))$
Initialize x_i using evenly spaced grid over $[0, n-1]$ with uniform noise $U(-0.5, 0.5)$
Velocities $v_i \leftarrow 0$
$pbest_i \leftarrow x_i$; $pbestval_i \leftarrow +\infty$
$gbest \leftarrow x_1$; $gbestval \leftarrow +\infty$
for $t \leftarrow 1$ to T **do**
  $w \leftarrow w_{max} - (w_{max} - w_{min}) \cdot \frac{t-1}{T-1}$
  for all particles $i \in \{1, \ldots, P\}$ **do**
    $idx \leftarrow \text{round}(\text{clip}(x_i, 0, n-1))$
    $victim \leftarrow visible\_victims[idx]$
    $dist_{norm} \leftarrow d(uav, victim)/\tau_d$
    $score_{norm} \leftarrow (score(victim)/10)^2$
    $conflict \leftarrow$ fraction of other AAVs intending $victim$ + safeguard penalty if $victim.id \in committed\_targets$
    $J \leftarrow \alpha \cdot dist_{norm} - \beta \cdot score_{norm} + \gamma \cdot conflict$
    if $J < pbestval_i$ **then**
      $pbestval_i \leftarrow J$; $pbest_i \leftarrow x_i$
    end if
    if $J < gbestval$ **then**
      $gbestval \leftarrow J$; $gbest \leftarrow x_i$
      reset stagnation counter
    end if
    sample $r_1, r_2 \sim U(0, 1)$
    $v_i \leftarrow wv_i + c_1 r_1 (pbest_i - x_i) + c_2 r_2 (gbest - x_i)$
    clamp v_i to $[-v_{max}, v_{max}]$
    $x_i \leftarrow \text{clip}(x_i + v_i, 0, n-1)$
  end for
  if stagnation counter $\geq \max(3, \lfloor T/3 \rfloor)$ **then**
    break ▷ Early stopping
  end if
end for
$idx^* \leftarrow \text{round}(\text{clip}(gbest, 0, n-1))$
return $visible\_victims[idx^*]$

where $d(C, p)$ is the Euclidean distance between the anchor target C and a neighboring victim p, and score(p) represents the victim's priority (2, 6, or 10, corresponding to the *green*, *orange*, and *gray* classes). The geometric evaluation radius is empirically set to $r_{\text{drop}} = 4.0$ spatial units (400 meters), adopting a safety margin relative to the actual effective radius of the supply delivery ($r_{\text{effective}} = 5.0$ spatial units, or 500 meters), to tolerate last-minute victim displacements without loss of coverage.

To decide whether the AAV needs to abandon its current target (C_{current}) in favor of a new candidate cluster (C_{new}), the agent computes the tactical area gain (ΔV) and the logistical distance gain (Δd), given by

$$\Delta V = V_{cluster}(C_{\text{new}}) - V_{cluster}(C_{\text{current}}) \tag{5}$$

$$\Delta d = d(\text{AAV}, C_{\text{current}}) - d(\text{AAV}, C_{\text{new}}) \tag{6}$$

The AAV trajectory change is only authorized if it satisfies a strict hysteresis transition rule, expressed by:

$$\text{Switch Target} \iff \Delta V > 0 \vee (\Delta V = 0 \wedge \Delta d > \tau_{dist}) \tag{7}$$

where $\tau_{dist} = 10.0$ acts as a spatial hysteresis threshold. The incorporation of this algorithmic constraint is of vital importance for the swarm's navigation stability, as it prevents the phenomenon of continuous route oscillation (*chattering effect*). Equation 7 ensures that a AAV will only abandon its current trajectory if the new cluster offers a strictly greater humanitarian impact ($\Delta V > 0$), or if, for an equivalent rescue gain, the new cluster is considerably closer, thereby optimizing the vehicle's response time and energy consumption. Note that any strictly positive humanitarian gain ($\Delta V > 0$) authorizes trajectory switching regardless of the distance penalty ($\Delta d < 0$). This design choice deliberately prioritizes impact maximization over fuel efficiency, which is appropriate in time-critical SAR scenarios where victim survival outweighs operational cost. However, for strictly fuel-constrained operations, a weighted condition such as $\Delta V > \lambda \cdot (-\Delta d)$ might be preferable in future iterations.

This dynamic is illustrated in Case 4 of Figure 1, which demonstrates the Layer 2 Spatial Tactical Evaluation. With the *Area-of-Effect* rescue capability enabled, AAV A identifies a cluster whose aggregated humanitarian impact surpasses that of an isolated critical target ($\Delta V > 0$). Consequently, the swarm executes a tactical trajectory shift toward the zone of greatest strategic value, effectively breaking the strict one-to-one rescue limit.

F. EVALUATED CONFIGURATIONS

To demonstrate the effect of the proposed refinement, we evaluated four configurations under the same simulation parameters: (i) Baseline: without WTA communication and without PSO; (ii) WTA: communication protocol from the baseline framework [9]; (iii) WTA–PSO without Layer 2: integration of local PSO-based intention optimization while preserving the original WTA conflict resolution; (iv) WTA–PSO with Layer 2: the complete proposed architecture, including both intention optimization and spatial clustering for area-of-effect rescues.

These four configurations are evaluated across three scenario scales (Table 3): reduced, baseline, and expanded.

G. EVALUATION METRICS

The following metrics are reported: (i) total rescue impact per generation (*Total_Value*), (ii) rescues per class, and (iii) distribution comparisons via *boxplots* (per generation and per seed). The total impact is defined by:

$$V_{\text{total}} = 2 \cdot N_{\text{Green}} + 6 \cdot N_{\text{Orange}} + 10 \cdot N_{\text{Gray}}.$$

where the weights (2, 6, 10) are assigned to each urgency level to reflect the increasing humanitarian priority of critical victims, following a non-linear scoring scheme previously established in [9]. Additionally, we report the total number of victims assisted per generation (*Total_Rescued*) to verify the combined effect of prioritization and spatial clustering on the success rate of the logistical fleet.

H. STATISTICAL ANALYSIS

Non-parametric tests are used to compare rescue distributions across classes and methods. For differences between classes (*green/orange/gray*), the Kruskal–Wallis test and, when appropriate, the Dunn–Bonferroni post-hoc test are applied. The effect size is reported according to [29] by:

$$\varepsilon^2 = \frac{H-k+1}{n-k},$$

where H is the Kruskal–Wallis statistic, k is the number of groups, and n is the total number of observations.

For the multi-seed comparison, is adopted the paired approach from the base framework [9]: the average *Total_Value* across generations per seed is computed and the Wilcoxon signed-rank test is applied to the increment Δ per seed to compare the approaches.

I. REPRODUCIBILITY

The simulations are executed in Python (3.x), using the NumPy, Pandas, Matplotlib, and SciPy libraries. A representative execution with seed $= 42$ and a multi-seed analysis comprising 50 independent runs (seeds 42–91). Additional sensitivity analyses utilize 10 seeds (42–51). The proposed modification preserves the mobility mechanisms and the WTA conflict resolution rules, ensuring full mathematical consistency with the foundations established in [9].

III. RESULTS

A. ANALYSIS OBJECTIVE AND NOMENCLATURE

The goal of this section is to quantify the impact of the proposed decision-making architecture. The primary comparison assesses the effectiveness of adding a lightweight PSO-based refinement to the local target choice (Layer 1), while fully preserving the WTA communication protocol and distributed allocation logic. Accordingly, the core comparison is WTA vs. WTA–PSO under the individual engagement mode (*single-payload drop*). The performance gain obtained by enabling Spatial Tactical Evaluation (Layer 2) is evaluated. In addition, a Baseline configuration (without WTA and without PSO) is included as a reference to highlight the overall effect of communication. The main robustness analysis in this section uses 50 seeds (42–91), whereas the sensitivity analyses (hyperparameters and variants) use 10 seeds (42–51).

To reflect the need for medical triage, three urgency classes are considered: *green* (low risk), *orange* (moderate risk), and *gray* (critical state), with scores (2, 6, 10) as listed in Table 4. The baseline scenario in this section uses $N_{\text{uav}} = 30$ AAVs and $N_{\text{victim}} = 170$ victims. As an additional scale-sensitivity analysis, a reduced scenario ($N_{\text{uav}} = 15$ and proportionally fewer victims) and an expanded scenario ($N_{\text{uav}} = 60$ and $N_{\text{victim}} = 340$, with 200 *green*, 80 *orange*, and 60 *gray*) are evaluated, while keeping the same scoring scheme and coordination logic.

Regarding the baseline configuration for the proposed WTA-PSO architecture, the dynamic budget is capped at a maximum of $P = 20$ particles and $I = 15$ iterations. The PSO hyperparameters are set with an inertia weight linearly decaying from $w_{\max} = 0.9$ to $w_{\min} = 0.4$, acceleration coefficients $c_1 = c_2 = 1.6$, and a velocity clamping of $v_{\max} = 2.0$. The early stopping mechanism is triggered if the global best value stagnates for 5 iterations. Finally, the multi-objective cost function weights are rigorously balanced at $(\alpha, \beta, \gamma) = (0.15, 1.4, 0.3)$.

TABLE 4. Urgency levels, displacement weights, quantities, and scores (baseline scenario).

Class (Urgency)	Color	Displacement weight t_c	Quantity	Score
Green (Low)	Green	0.25	100	2
Orange (Medium)	Orange	0.50	40	6
Gray (High)	Gray	0.15	30	10

B. REPRESENTATIVE RUN (SEED = 42): CLASS-WISE BEHAVIOR AND LOGISTICAL STABILITY

A representative run with seed $= 42$ is presented, which is also used as a consistency check of the environment. Figure 4 shows the distribution of rescues per generation, stratified by urgency class and comparing Baseline, WTA, and WTA–PSO.

Under the Baseline setting, rescues are strongly concentrated in the *green* class (mean 19.80, standard deviation 2.16), with the *orange* class presenting a mean of 6.04 (standard deviation 1.96) and the *gray* class registering the lowest service levels (mean 4.16, standard deviation 1.62). With WTA, the dominance of *green* decreases (mean 15.68, standard deviation 2.57) and rescues of *gray* victims increase (mean 6.80, standard deviation 2.09), while the *orange* class remains close (mean 6.46, standard deviation 2.11). With WTA–PSO, we observe a sustained strategic shift toward the *gray* class (mean 8.02, standard deviation 2.38), with *green* at 16.04 (standard deviation 2.85) and *orange* reaching 5.70 (standard deviation 1.92). In practical terms, WTA–PSO reallocates logistical effort from lower-urgency victims to critical targets (*gray*), while keeping WTA's coordination and conflict-resolution logic unchanged.

Across all three configurations, the class effect is statistically significant (Kruskal–Wallis, $p < 10^{-20}$). The global effect sizes were $\varepsilon^2 = 0.7304$ (Baseline), $\varepsilon^2 = 0.6635$ (WTA), and $\varepsilon^2 = 0.7219$ (WTA–PSO), suggesting that coordination mechanisms make the rescue distribution less accidentally concentrated in a single class. Dunn–Bonferroni post hoc tests confirm significant differences among all classes in the Baseline (all pairwise comparisons with

$p_{\mathrm{adj}} \leq 0.007$) and in WTA–PSO (all pairwise comparisons with $p_{\mathrm{adj}} \leq 0.003$); under WTA, the *orange* and *gray* distributions do not differ significantly after correction ($p_{\mathrm{adj}} = 1.0$), which is fully consistent with the proximity of the means observed for this method.

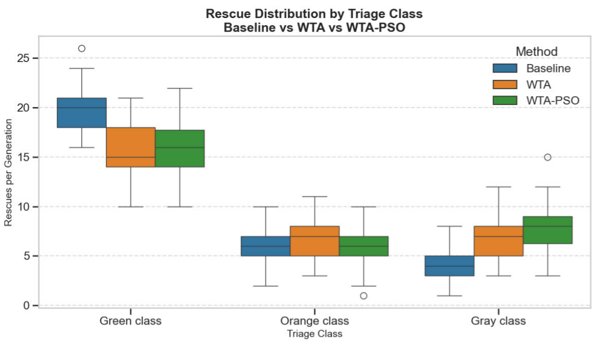

FIGURE 4. Rescue distribution by class for the representative run (seed = 42), comparing Baseline (no WTA / no PSO), WTA, and WTA–PSO. WTA–PSO reduces incidental service in the *green* class and increases critical rescues in the *gray* class, while keeping *orange*-class service at stable levels.

To isolate the central Layer 1 hypothesis, it is verified if the WTA–PSO gain can be explained by an increase in the total service volume (rather than by intelligent prioritization). Figure 5 reports the total number of deliveries per generation for WTA and WTA–PSO in the same run (seed = 42). Both methods operate close to the scenario's logistical limit in most generations, with only minor differences in volume (mean 28.94 for WTA and 29.76 for WTA–PSO). Therefore, the tactical impact gain of WTA–PSO (Layer 1) is purely associated with triage optimization across classes, not with an increase in the raw delivery rate.

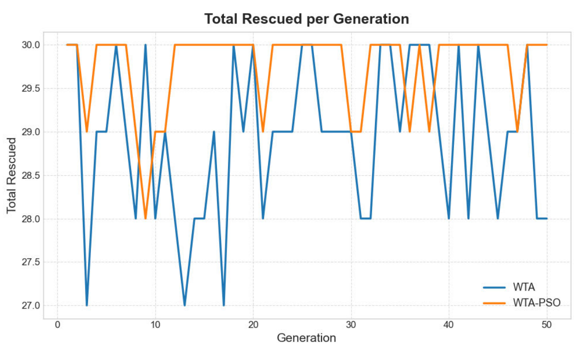

FIGURE 5. Total rescues per generation in the representative run (seed = 42), comparing WTA and WTA–PSO. The curves remain close and operate near the scenario's logistical ceiling, indicating that the humanitarian gain of WTA–PSO does not stem from a higher raw delivery volume, but from strict prioritization of critically endangered lives.

C. MULTI-SEED ROBUSTNESS (50 INDEPENDENT RUNS): GAIN IN HUMANITARIAN IMPACT

To assess robustness, a multi-seed analysis is conducted with 50 independent runs (seeds 42–91), each with 50 generations. Considering all generations and seeds, the mean rescue impact (*Total_Value*) increases from 120.70 (standard deviation 16.20) in the Baseline to 145.52 (standard deviation 20.78) with WTA, and to 153.28 (standard deviation 19.74) with WTA–PSO.

In relative terms, WTA–PSO improves *Total_Value* by approximately 5.33% compared to WTA. At the same time, the number of victims served remains very close: mean *Total_Rescued* of 29.48 (WTA) and 29.73 (WTA–PSO), while the Baseline keeps 30.00 by construction due to the scenario mechanics. These results support the central hypothesis: the proposed PSO does not reduce overall service capacity, but maximizes the impact of each flight by favoring higher-urgency victims.

Figure 6 presents the evolution of *Total_Value* as mean $\pm$ standard deviation over the 50 seeds.It is observed that WTA–PSO tends to remain above WTA throughout most generations, indicating a moderate yet consistent gain.

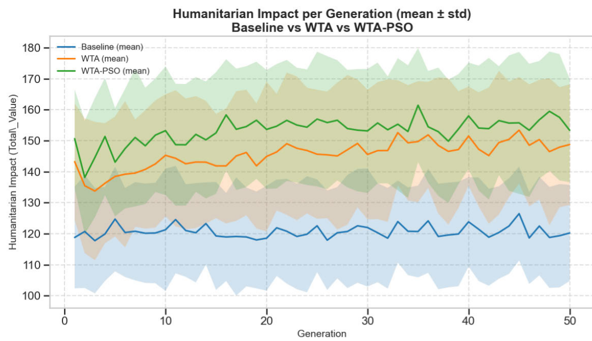

FIGURE 6. Total humanitarian impact per generation (*Total_Value*) shown as mean ± standard deviation over 50 seeds (42–91). WTA–PSO tends to achieve higher mean values than WTA while maintaining a similar logistical rate.

For a conservative and directly paired comparison, it is computed, for each *seed*, the mean humanitarian impact (*Total_Value*) across generations. Figure 7 shows the distribution of these per-seed means, with medians of 120.96 (Baseline), 147.18 (WTA), and 152.86 (WTA–PSO). A Kruskal–Wallis test on these means confirmed highly significant differences across methods ($H = 103.50$, $p < 10^{-10}$). Because the runs share the same spatial initializations (*seeds*), a paired Wilcoxon test is applied to the per-seed impact increment (Δ). For WTA–PSO vs. WTA, it is obtained $W = 234.5$ and $p = 1.00 \times 10^{-4}$ (two-sided), with a mean gain of $\Delta = 7.76$ and a 95% confidence interval (CI) of [4.47, 11.02]; moreover, 74% of the *seeds* yield $\Delta > 0$. Direct comparisons against the Baseline are also highly significant ($W = 0$; $p \ll 10^{-9}$), demonstrating the superiority of the proposed architecture in maximizing rescue impact.

Finally, Figure 8 summarizes the mean number of rescues by urgency class aggregated over the 50 *seeds*, corroborating the swarm shift toward critical targets.

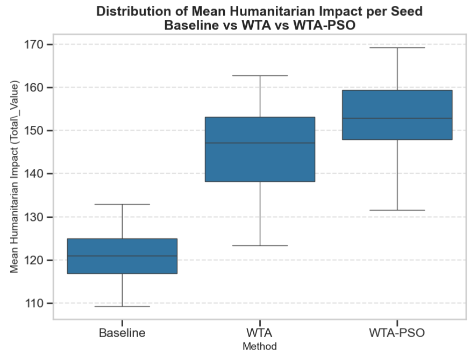

FIGURE 7. Distribution of the per-seed mean humanitarian impact (*Total_Value*) over 50 *seeds* (42–91). WTA–PSO exhibits a higher median than WTA, reinforcing the consistency of prioritizing critical victims even under severe stochastic uncertainty.

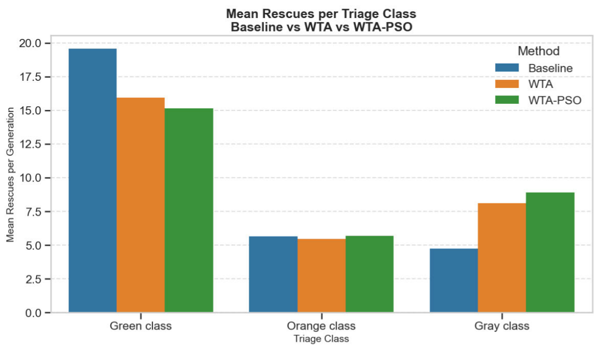

FIGURE 8. Mean rescues by urgency class aggregated over 50 *seeds* (42–91), comparing Baseline, WTA, and WTA–PSO. The plot highlights the effort reallocation induced by WTA–PSO, which reduces low-risk rescues (*green*) to maximize service to critically endangered victims (*gray*).

D. SENSITIVITY TO PSO HYPERPARAMETERS

The sensitivity of the proposed method is evaluated to operational hyperparameter choices in the baseline scenario, while keeping the environment topology, WTA conflict-resolution protocol, and tactical cost function (Equation 2) fixed. Due to the high computational cost inherent to parameter sweeps in a dynamic multi-agent simulator, this exploratory stage is conducted with a representative subset of 10 independent runs (*seeds* 42–51). While the core architectural validation was rigorously evaluated over 50 seeds, this 10-seed subset proved sufficient to observe the general stability trends of the hyperparameters without incurring prohibitive computational times.

As the primary effectiveness metric, it is notad the per-seed humanitarian-impact gain relative to the standalone WTA performance: $\Delta Total\_Value = \overline{V}_{\text{config}} - \overline{V}_{\text{WTA}}$. The PSO computational budget is strictly controlled by the maximum particle-swarm size (P) and the iteration limit (I). With I fixed at 15 to ensure real-time feasibility, three particle caps are compared: P5_I15, P10_I15, and P20_I15. In addition, extreme variations of the conflict-penalty term γ (GAMMA_0, removing predictive avoidance; and GAMMA_6 are tested, imposing severe aversion to route collisions).

Figure 9 and Table 5 summarize the median, mean, and win rate (*Positive_Rate*) of the increment Δ. It is observad a classic *trade-off* between tactical robustness and processing cost: P5_I15 is computationally lightweight, but its reduced sampling makes it less effective at identifying the best victim; in contrast, P10_I15 yields the highest win-rate consistency (80%), albeit with a lower impact ceiling. The 20-particle cap (P20_I15) is assumed, anchored to the dynamic scaling rule $\min(20, |\mathcal{V}_{vis}|)$, as the default configuration of the architecture. This calibration achieves the best balance point: it provides the highest potential for tactical gains in complex clusters (maximum median of 7.40), while the adaptive swarm-sizing rule (Layer 1) mitigates computational waste and stabilizes runtime.

Furthermore, the computational cost analysis (Table 5) demonstrates the lightweight nature of the algorithm. The mean total execution time for the default P20_I15 configuration is 45.10 seconds per complete simulation run. Given the swarm size and the finite horizon of iterations, this translates to a fraction of a millisecond per individual AAV decision step, validating its feasibility for real-time embedded applications on hardware with limited processing power.

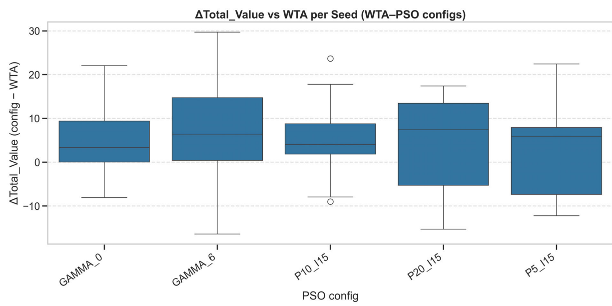

FIGURE 9. Sensitivity of the WTA–PSO architecture in the exploratory baseline scenario (10 *seeds*). The box plot shows the distribution of Δ*Total_Value* for different computational budgets (P, I) and variations in the conflict-penalty weight (γ).

TABLE 5. Statistical summary of the PSO hyperparameter sweep in the baseline scenario (10 *seeds*).

Configuration	Median	Mean	Standard deviation	Positive_Rate	Mean time (s)
P5_I15	5.92	2.54	10.94	0.70	36.62
P10_I15	4.00	5.41	10.08	0.80	41.83
P20_I15	7.40	4.40	11.73	0.70	45.10
GAMMA_0	3.32	5.97	9.61	0.70	49.85
GAMMA_6	6.38	5.74	14.47	0.70	48.40

E. SCALE SENSITIVITY (0.5× AND 2×): BEHAVIOR UNDER REDUCED AND ENLARGED FLEETS

To verify the consistency of the tactical approach under critical variations of the Search-and-Rescue environment, the multi-seed analysis is repeated (50 independent initializations) in the reduced ($N_{\text{uav}} = 15$, $N_{\text{victim}} = 85$) and enlarged ($N_{\text{uav}} = 60$, $N_{\text{victim}} = 340$) scenarios, while strictly preserving the same victim demographic proportions. The conservative per-*seed* analysis and paired Wilcoxon tests are considered.

Reduced scenario ($N_{\text{uav}} = 15$). The global mean Humanitarian Impact (*Total_Value*) was 59.30 (Baseline), 70.02 (WTA), and 75.58 (WTA–PSO), with the total number of deliveries (*Total_Rescued*) stabilized around the fleet's severe logistical ceiling (15.00). In relative terms, the tactical gain of WTA–PSO rises to approximately +7.94% over WTA. The paired per-*seed* comparison yields $W = 157.0$ and a p-value of 6.35×10^{-7}, indicating extremely high significance. This result suggests that, with smaller fleets and highly scarce resources, the geometric myopia of conventional engagement heuristics strongly penalizes the mission, making proactive triage optimization (*Particle Swarm*) even more critical to overall survivability.

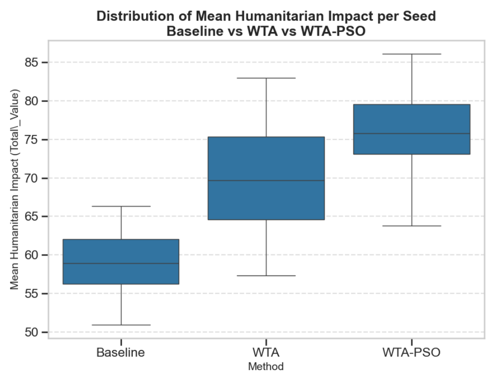

FIGURE 10. Distribution of the per-*seed* mean *Total_Value* in the high-scarcity reduced scenario ($N_{\text{uav}} = 15$, 50 *seeds*). The performance jump of WTA–PSO corroborates the effectiveness of triage under restrictive conditions.

Enlarged scenario ($N_{\text{uav}} = 60$). In a high-density environment with many agents and victims, spatial competition intensifies substantially. The medians of raw service remain pinned to the limit (60 rescues) for the coordinated methods. The global mean *Total_Value* was 243.14 (Baseline), 287.79 (WTA), and 295.45 (WTA–PSO). The proposed method maintains a consistent supplementary gain of +2.66% over WTA. The Wilcoxon test across the 50 *seeds* ($W = 341.0$, $p = 0.0036$) confirms that the advantage is statistically robust, rejecting the null hypothesis. This indicates that WTA–PSO mitigates the negotiation complexity and trajectory conflicts inherent to large (supersaturated) swarms, preserving prioritization integrity where conventional algorithms degrade due to redundant intentions.

F. LAYER 2: SPATIAL GROUPING (AoE) IMPACT ON MULTIPLE RESCUES

The results of the previous sections assume a strict one-to-one service regime. To evaluate the fleet's triage and prioritization capability (Layer 1) in isolation, the simulation environment included an algorithmic constraint (an engagement *lock*) that forced each payload to benefit exclusively a single victim per

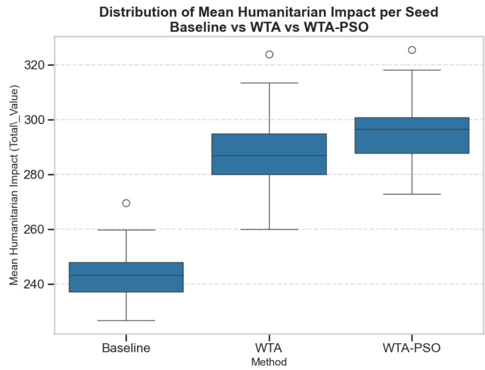

FIGURE 11. Distribution of the per-*seed* mean *Total_Value* in the supersaturated enlarged scenario ($N_{\text{uav}} = 60$, 50 *seeds*). The sustained statistical superiority demonstrates the swarm architecture's scalability under massive coordination complexity.

flight, physically limiting mission efficiency to a ceiling of N_{uav} rescues per generation.

In this stage, this restrictive lock is removed and enable the physical *Area-of-Effect* (AoE) delivery capability, allowing a single payload to serve all victims simultaneously within the effective radius (5.0 spatial units, equivalent to 500 meters). To ensure a fair comparison, this multi-rescue mechanic is enabled for all three approaches (Baseline, WTA, and WTA–PSO). However, the second proposed architectural layer (Spatial Tactical Evaluation via *Clustering*, based on Equation 4) operates jointly with PSO to proactively steer the swarm toward the most valuable clusters.

When running the fleet in the baseline scenario ($N_{\text{uav}} = 30$) over 50 independent spatial initializations (seeds 42–91) without the single-engagement lock, all methods break the strict logistical ceiling of 30 rescues, benefiting from the disaster's population density. The Baseline, acting in an uncoordinated manner, achieves an overall mean of 43.10 incidental rescues (Impact of 190.28). The traditional WTA, coordinating targets by proximity, reaches 44.06 rescues (Impact of 228.28).

The superiority of the two-layer architecture, however, becomes unequivocal with WTA–PSO. Guided by cluster-aware geometric perception and triage optimization, the method rises to an average of 44.80 victims served using exactly the same 30 vehicles. More important than the raw volume is the jump in overall humanitarian impact (*Total_Value*), which reaches a mean of 239.15 points—an expressive increase of approximately +56% compared to the 153.28 achieved by the same fleet in the locked (one-to-one) mode—while also securing a sustained advantage of +10.87 points in direct comparison with lock-free WTA.

To demonstrate that the Layer 2 efficiency jump of WTA–PSO is not merely the result of incidental rescues in dense low-risk areas, the multi-seed demographic breakdown of area rescues is used. Under the Baseline, service is strongly concentrated in the *green* class (mean 24.87), with *orange* at 10.44 and *gray* accounting for the smallest fraction (mean

7.79). WTA mitigates part of this disparity, yielding means of 20.90 (*green*), 11.27 (*orange*), and 11.89 (*gray*). The true cooperative tactical gain emerges with WTA–PSO: the fleet actively steers its coverage zones toward critical patients, reaching a substantial mean of 12.91 simultaneous rescues in the *gray* class, in addition to 11.57 in *orange* and 20.32 in *green*.

As in the strict regime, the effect of collaborative intelligence in the area-of-effect regime is reflected in high global statistical significance (Kruskal–Wallis, $H = 105.26$, $p < 10^{-10}$). Moreover, the paired Wilcoxon analysis across the 50 *seeds* confirms the systematic gain of WTA–PSO over WTA in this spatial regime ($W = 236.0$; $p = 5.37 \times 10^{-5}$), with a mean increment of $\Delta = 10.87$ points and a win rate of 70%.

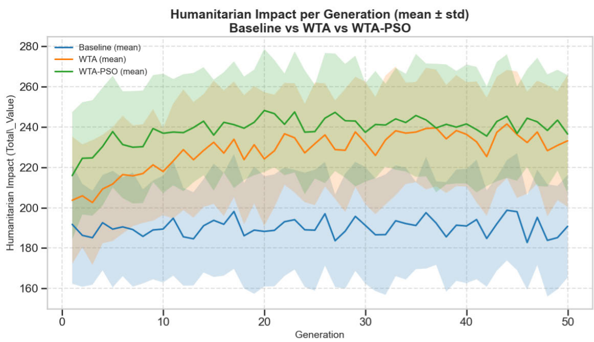

FIGURE 12. Humanitarian-Impact evolution (mean ± standard deviation) with Spatial Clustering (Layer 2) enabled over 50 independent *seeds*. Removing the single-rescue constraint allows WTA–PSO to reach and sustain higher levels, consistently outperforming the standard approaches.

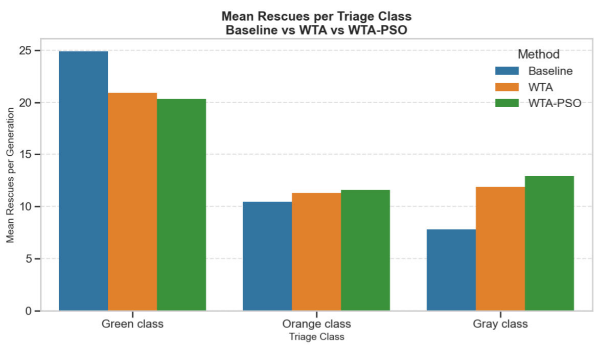

FIGURE 13. Mean rescues by urgency class under Layer 2 (50 *seeds*). The impact jump of WTA–PSO is explained by its ability to actively pursue high-value zones, converting constrained logistical resources into the maximum rate of critical rescues (*gray*).

This phenomenon validates the robustness of this proposed model in real-world disasters: while standard methods rely on chance or simple proximity to produce multi-victim rescues, WTA–PSO may forgo isolated critical victims if it identifies—via the navigation hysteresis mechanism (Equation 7)—an adjacent delivery zone (*drop zone*) that mathematically maximizes the collective survival rate.

IV. DISCUSSION

A. LIMITATIONS AND FUTURE DIRECTIONS

Although the proposed architecture (WTA-PSO and Spatial Clustering) delivers consistent and statistically significant gains across all evaluated operating scales ($N_{\text{uav}} \in \{15, 30, 60\}$), limitations of the model can be highlighted. The rigorous multi-seed analysis shows that triage intelligence is universally beneficial ($p < 0.01$ in all paired tests); however, its relative impact is markedly larger under regimes of extreme logistical scarcity. In supersaturated environments, the high vehicle density naturally reduces the swarm's tactical maneuvering margin, constraining the ceiling of percentage gains. In addition, within the Layer 2 context (area deliveries), the model assumes that all victims within the effective radius fully receive the payload benefit, which in practice may be affected by disaster topography, debris occlusions, or adverse weather conditions that are not represented in the current 2D kinematic simulation, which also abstracts away hardware-specific constraints such as battery consumption and aerodynamic disturbances.

Regarding communication, while the architecture is specifically designed for disaster zones where terrestrial networks have collapsed, the current testbed isolates the baseline performance of the tactical algorithm by modeling the local inter-vehicle (drone-to-drone) communication under idealized conditions, assuming instantaneous intention processing and noise-free sensing. This abstraction is adopted to validate the core WTA–PSO mechanics without the confounding variables of signal degradation. Moreover, the weights of the tactical cost function (distance, urgency, and conflict penalty) are set statically, without an exhaustive and systematic hyperparameter calibration aimed at real-time adaptation.

As future work, the operational realism can be increased by subjecting the local *mesh* network to realistic constraints, such as variable delays, limited bandwidth, and dynamic topology disruptions. From a humanitarian perspective, it is possible to introduce temporal degradation of victims' health states, where urgency classes evolve progressively (e.g., transition from *orange* to *gray*) as rescue time increases. Finally, the *Reinforcement Learning* approaches coupled with PSO can be explored to enable AAVs to autonomously calibrate the tactical-evaluation weights based on the perceived level of chaos in the environment, maximizing swarm resilience under extreme uncertainty scenarios. Additionally, future studies will prioritize comparing the proposed architecture against other recognized multi-agent task allocation baselines, such as the Consensus-Based Bundle Algorithm (CBBA), before advancing toward physical validation with real AAV swarms.

V. CONCLUSION

This work proposed a two-layer tactical decision architecture for swarms of Autonomous Aerial Vehicles (AAVs) operating in Search-and-Rescue (SAR) missions under severe logistical constraints. In Layer 1, the proposed WTA–PSO algorithm replaces the geometric myopia of conventional engagement heuristics with a lightweight multi-criteria optimization. Rigorous multi-seed tests (50 independent initializations) show that the method actively reallocates fleet resources: vehicles no longer serve low-risk victims (*green*) merely due to spatial opportunism, and instead prioritize targets in critical condition (*gray*). This behavior yields a consistent and statistically significant increase in global Humanitarian Impact across all evaluated operating scales ($N_{\text{uav}} \in \{15, 30, 60\}$), while operating at the fleet's payload-capacity limit without sacrificing the raw service volume.

The true jump in tactical efficiency, however, is demonstrated in Layer 2 of the architecture. By enabling spatial clustering perception (area deliveries) coupled with a navigation-hysteresis function, the swarm transcends strict one-to-one allocation. WTA–PSO proves capable of actively seeking and prioritizing high-value *drop zones*, breaking the fleet's mechanical logistical ceiling. In the baseline scenario with 30 vehicles, the method increases from 30 to a mean of 44.80 simultaneous rescues, producing an expressive impact gain (+56% relative to the locked regime) and securing an unequivocal statistical superiority of $p < 10^{-4}$ over the traditional WTA protocol.

In summary, the results show that integrating low-computational-cost metaheuristics (such as PSO) with distributed conflict-resolution protocols (WTA) yields a highly resilient disaster-response system. Targeted injections of cooperative intelligence at the local flight-intention generation stage are sufficient to transform robotic fleets that are severely constrained by *payload* into survivability-maximizing tools. These simulation-based findings suggest promising directions for real-world deployment, pending validation under realistic communication constraints, sensor noise, and dynamic field conditions.

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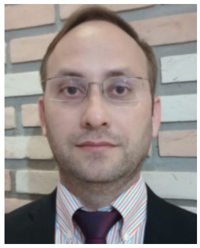

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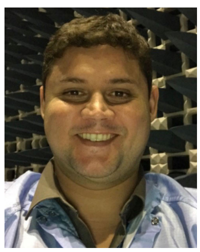

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